Gas Powered Cycles

For (cyclic) engines, the thermodynamic cycle they operate on is called a power cycle. There are two types of power cycles, gas power cycles, when the working fluid stays as a gas, and vapour powered cycles, when the working fluid changes phase to a liquid at one point in the cycle. This chapter focuses on gas powered cycles.

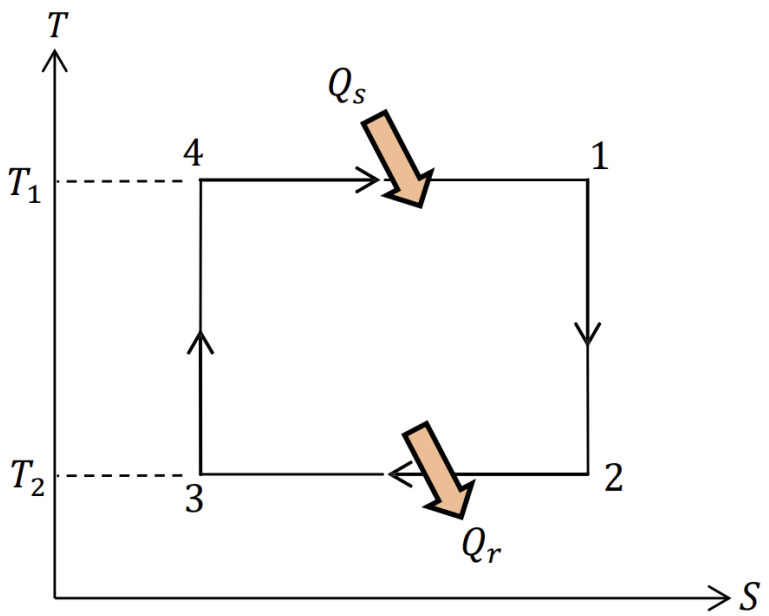
There are 3 main types gas powered cycles, Otto cycle (for spark ignition petrol engines), Diesel cycle (compression ignition petrol engines), and the Dual Combustion cycle.

The Carnot Cycle

The Carnot cycle is an ideal process of a maximum efficiency engine. Recall that the second law of thermodynamics means that a 100% efficient engine is not possible, as some heat has to be rejected by the Kelvin-Planck statement.

However, reversible processes can maximise the efficiency of these engines, which is what the Carnot cycle uses.

The Carnot cycle has 4 processes:



4 → 1: reversible isothermal heat supply/expansion

The system is allowed to gain heat from a hot reservoir, and will expand isothermally.

1 → 2: isentropic expansion

After a while, the system is connected to an insulator, but will continue to expand, until the working fluid cools to the same temp as the cold reservoir.

2 → 3: reversible isothermal heat rejection/ compression

The system is then connected to a cold reservoir, where it causes it to compress isothermally.

3 → 4: isentropic compression

the system is then connected to an insulator, where it continues to compress until the working fluid heats up to the same temp as the hot reservoir.

To determine the thermal efficiency of a cycle, we use

$$\eta = \frac{\text{Net work}}{\text{Heat Supplied}} = \frac{Q_s - Q_r}{Q_s} = 1 - \frac{Q_r}{Q_s} = 1 - \frac{T_2(S_2 - S_3)}{T_1(S_1 - S_4)}$$

As $S_2 - S_3 = S_1 - S_4$ in a Carnot cycle, the efficiency is

$$\eta = 1 - \frac{T_2}{T_1}$$

where T2 is the temperature of the cold reservoir, and T1 is the temperature of the hot reservoir.

However, the Carnot cycle has some limitations:

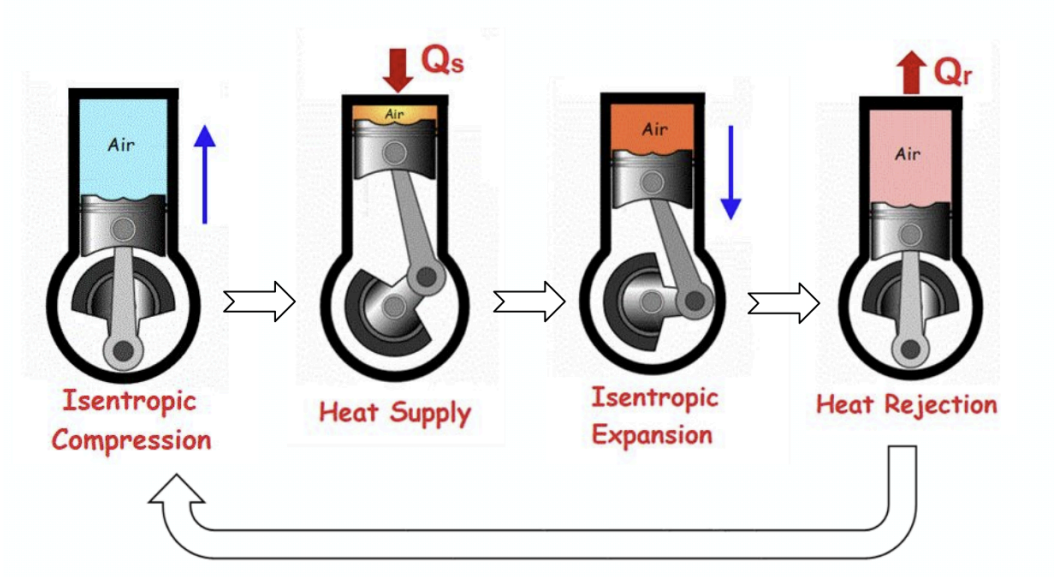
It is impractical for a system to gain or lose heat at a constant temperature, and is more practical to heat a gas at an (approximately) constant pressure or volume.

The Carnot cycle also has a low work ratio, meaning it outputs a low amount of work for the work put in.

$$\text{Work Ratio} = \frac{\text{Net Work}}{\text{Gross Work}} = \frac{W_o - W_i}{W_o}$$

Assumptions & Definitions of Gas Powered Engines

Actual cycles in gas powered engines are much more complicated. For example, a four stroke internal combustion engine undergoes the following strokes, or processes:



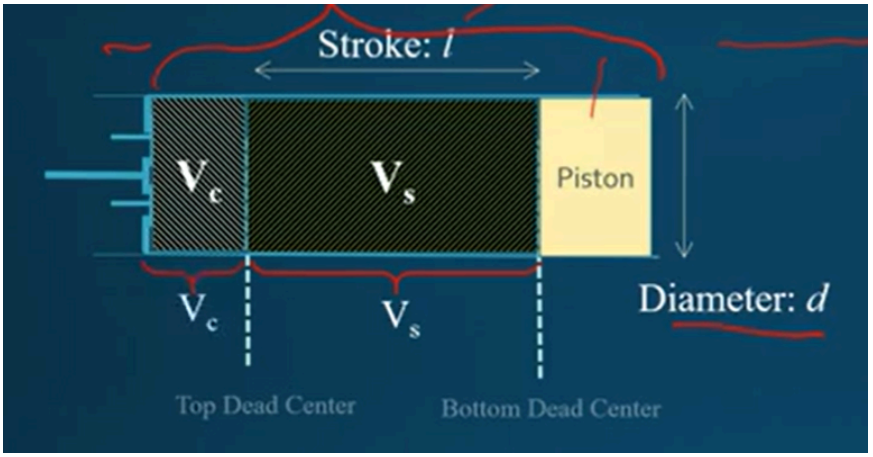
- Intake Stroke
- Compression Stroke
- Power Stroke
- Expansion Stroke

As these are harder to analyse due to more variables being involved, analysing air standard cycles makes it easier to analyse and optimise engines.

Hence, when we analyse air standard cycles, we make the assumptions that

- the cycle is a complete thermodynamic cycle
  - combustion is modelled as a heat addition process, aka the chemical reaction is not considered
- the working substance is air, as an ideal gas with constant specific heat capacity at 25C.
- all processes are internally reversible in the cycle.

All of the air standard cycles take place in a reciprocating engine, or a piston engine. The piston then sweeps along a length l in the cylinder.



Hence, the piston sweeps across a volume of

$$V_s = \frac{\pi d^2 l}{4}$$

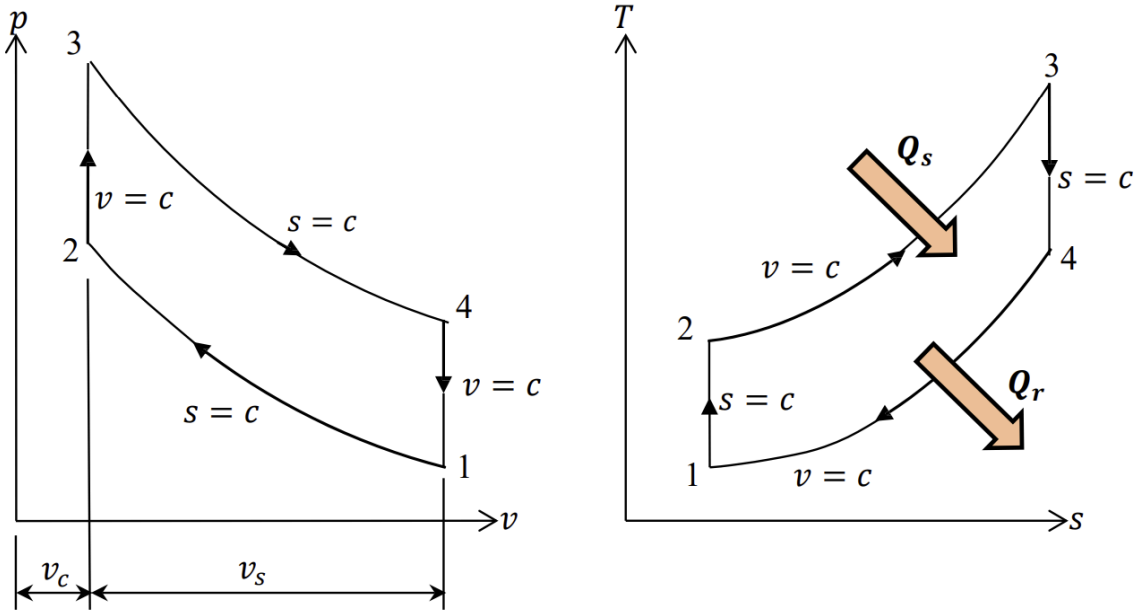
and if the clearance volume, V_c can be found, the compression ratio can be found as

$$r_v = \frac{V_c + V_s}{V_c} = 1 + \frac{V_s}{V_c}$$

The Otto Cycle

The Otto cycle is meant to approximate the petrol engine cycle, operating on the principle of heat supply and rejection under constant volume. Typically, the Otto cycle works with an inlet and outlet, as well as a spark plug, approximating petrol engines.

Shown is the PV and TS diagrams of the Otto cycle:



1 → 2: isentropic compression

The inlet allows some gas to enter the cylinder and the piston compresses the gas adiabatically.

2 → 3: isochoric heat supply

The spark plug is activated and causes the gas to combust, supplying heat to the system.

3 → 4: isentropic expansion

The high pressure and temperature air expands and causes the piston to move, causing a work output

4 → 1: isochoric heat rejection

The heat of the gas is lost to the environment and is allowed to flow out of the outlet afterwards

Each of these processes (isentropic, isochoric) can have their initial and final states calculated as usual. To relate them to the compression ratio, for example for 1 → 2:

$$r_v = \frac{v_s + v_c}{v_c} = \frac{V_1}{V_2}$$

which can be used for finding the pressure and temperature ratio.

The efficiency of the cycle can be found as

$$\eta = \frac{\Sigma W}{Q_s}$$

as said earlier. As the Otto cycle is a complete thermodynamic cycle,

$$\Sigma Q = \Sigma W$$
$$\Sigma W = Q_s - Q_r$$

Hence

$$\eta = \frac{Q_s - Q_r}{Q_s} = 1 - \frac{Q_r}{Q_s}$$

As the heat rejection/supply processes are isochoric,

$$\eta = 1 - \frac{\cancel{mC_v}(T_4 - T_1)}{\cancel{mC_v}(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

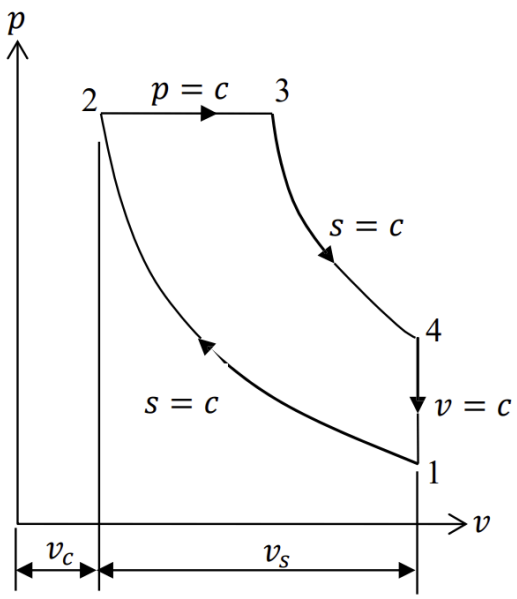
expressing the bottom in terms of T4 and T1 with adiabatic equations of the temperature ratio:

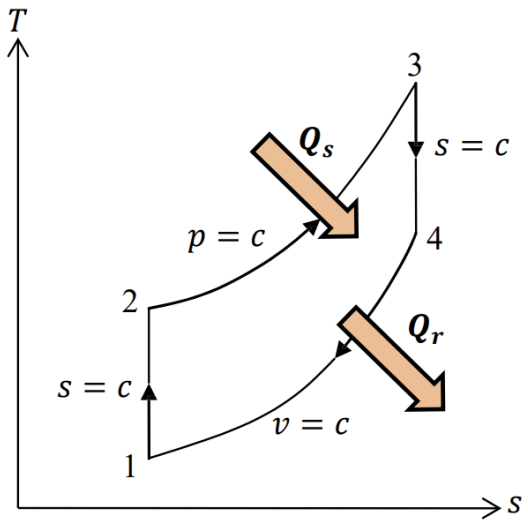
$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$
$$= 1 - \frac{T_4 - T_1}{T_4 r_v^{\gamma-1} - T_1 r_v^{\gamma-1}}$$
$$= 1 - \frac{T_4 - T_1}{(T_4 - T_1) r_v^{\gamma-1}}$$
$$\eta = 1 - \frac{1}{r_v^{\gamma-1}} = 1 - r_v^{1-\gamma}$$

Notice the efficiency is only affected by the compression ratio.

The Diesel Cycle

The Diesel Cycle can be thought of as a modified Otto cycle that supplies heat at a constant pressure. This is meant to approximate diesel engines, which operate at low speeds. Usually, diesel engines have a fuel injector instead of a spark plug for combustion.





1 → 2: isentropic compression

The inlet allows some gas to enter the cylinder and the piston compresses the gas adiabatically.

2 → 3: isobaric heat supply

Fuel is injected at the end of the stroke compression stroke and causes the gas to combust, supplying heat to the system. The piston is allowed to move freely, causing the cylinder pressure to stay constant, and the volume to increase slightly. Hence, some work is done at this process.

3 → 4: isentropic expansion

The high pressure and temperature air expands and causes the piston to move, causing a work output

4 → 1: isochoric heat rejection

The heat of the gas is lost to the environment and is allowed to flow out of the outlet afterwards

One quantity of the diesel cycle is the cut off ratio, defined as:

$$\beta = \frac{V_3}{V_2}, \text{ where } 2 \rightarrow 3 \text{ is the isobaric heat supply process.}$$

From the isobaric equation of state,

$$\frac{V_2}{T_2} = \frac{V_3}{T_3} \Rightarrow \frac{T_3}{T_2} = \frac{V_3}{V_2} = \beta$$

The cut off ratio is useful for the isentropic expansion, as

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} = \left(\frac{V_3}{V_2} \times \frac{V_2}{V_4}\right)^{\gamma-1} = \left(\beta \frac{1}{r_v}\right)^{\gamma-1}$$
$$= \left(\frac{\beta}{r_v}\right)^{\gamma-1}$$

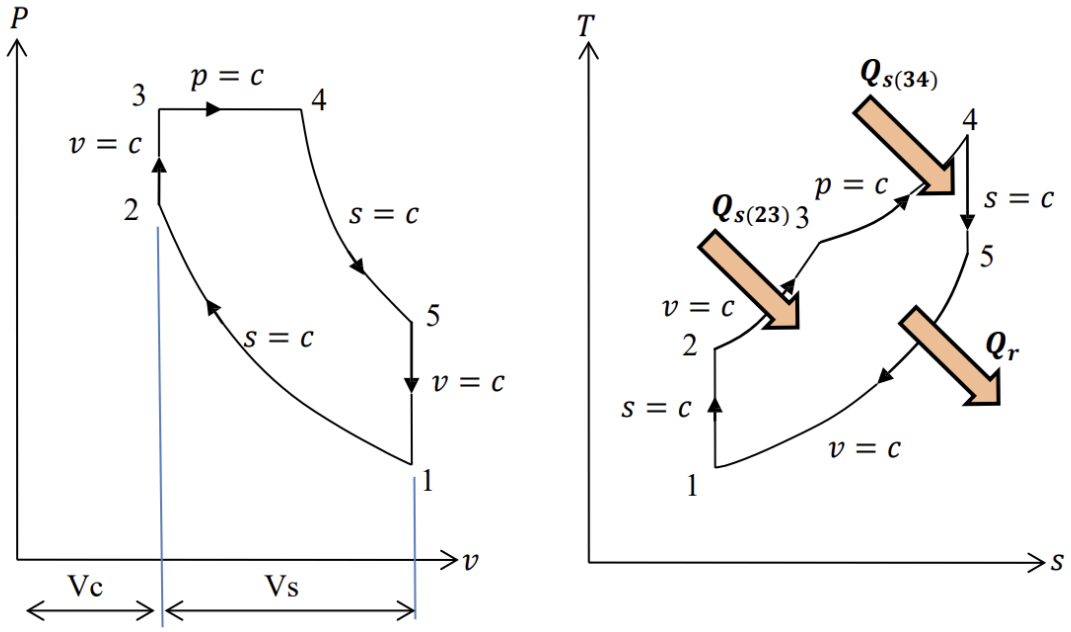
For the diesel cycle, the thermal efficiency is

$$\eta = 1 - \frac{\beta^\gamma - 1}{\gamma(\beta - 1) r_v^{\gamma-1}}$$

Notice the efficiency is dependent on the compression ratio, as well as the cut off ratio, which is dependent on the amount of heat added.

Dual Combustion Cycle

The Dual Combustion cycle can be thought of as a combination of the Otto and Diesel Cycle, as the heat is supplied under two processes, first under an isochoric process, then under an isobaric process. This cycle is also meant to approximate the high speed diesel engine.



1 → 2: isentropic compression

The inlet allows some gas to enter the cylinder and the piston compresses the gas adiabatically.

2 → 3: isochoric heat supply

Atomised diesel fuel is injected, causing the gas to combust spontaneously, causing heat to be supplied under constant volume first. Fuel is still being supplied past state 3

3 → 4: isobaric heat supply

As the combustion continues, the piston is allowed to move freely, causing the cylinder pressure to stay constant as it continued to combust, and the volume to increase slightly. Hence, some work is done at this process.

4 → 5: isentropic expansion

The high pressure and temperature air expands and causes the piston to move, causing a work output

5 → 1: isochoric heat rejection

The heat of the gas is lost to the environment and is allowed to flow out of the outlet afterwards

For the dual cycle, one quantity is the heat addition pressure ratio, given as

$$k = \frac{P_3}{P_2}, \text{ where } 2 \rightarrow 3 \text{ is the isochoric heat supply process.}$$

which is useful for finding the temperature in process 2 → 3.

The thermal efficiency can be found as

$$\eta = 1 - \frac{k\beta^{\gamma} - 1}{r_v^{\gamma-1}\left((k-1) + \gamma(k\beta - 1)\right)}$$

However, it is more practical and easier to calculate efficiency with the general formula of

$$\eta = 1 - \frac{Q_r}{Q_s}$$